Attacks on LLMs

1) Vapnik's SL (GD)

2) LLMs

4) Riem. Geom.

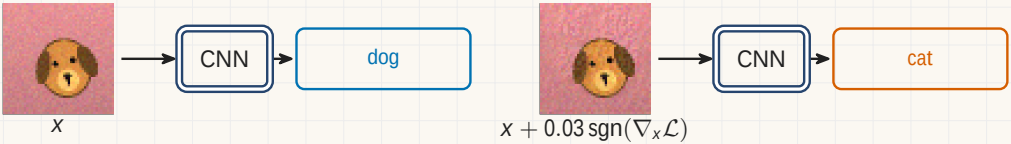
Weekend 4, Saturday morning

Carlos Cotrini, Jesus Solano

CAS Building ML/AI Applications, ETH Zurich

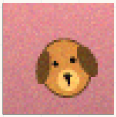
Saturday 24 October 2026

From a dog to a request



1. This morning: one step of 0.03 on every pixel, invisible to us, and the network says cat.

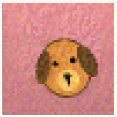
From a dog to a request



x

CNN

dog



$x + 0.03\,\mathrm{sgn}(\nabla_x \mathcal{L})$

CNN

cat

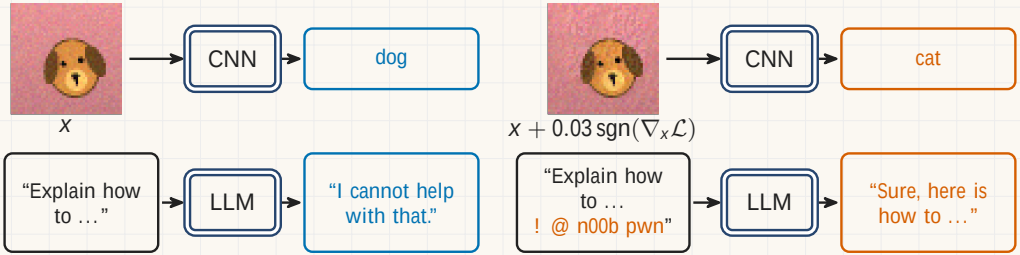
“Explain how to …”

LLM

“I cannot help with that.”

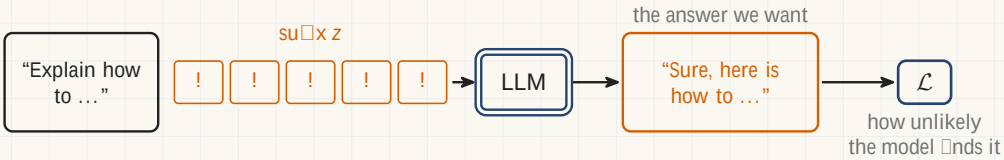
1. This morning: one step of 0.03 on every pixel, invisible to us, and the network says cat.
2. A chat model is trained to decline a harmful request.

From a dog to a request



1. This morning: one step of 0.03 on every pixel, invisible to us, and the network says cat.
2. A chat model is trained to decline a harmful request.
3. Add a few words chosen by the attacker, a su□x, and it complies.

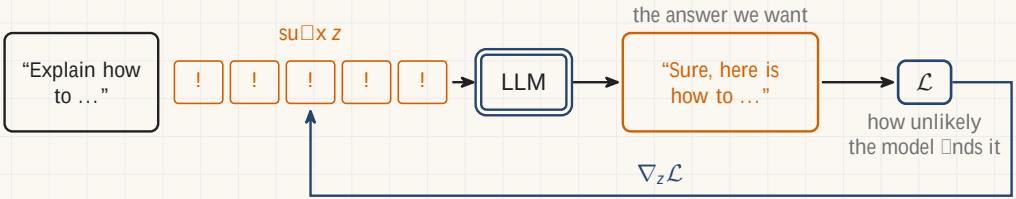
Finding the su□x



1. Aim for an answer that begins “Sure, here is”: once begun, the model keeps going.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

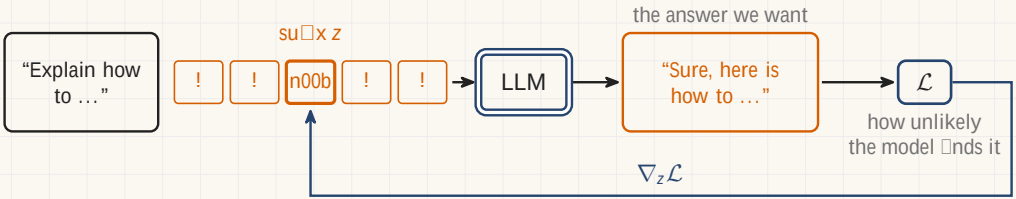
Finding the su□x



1. Aim for an answer that begins “Sure, here is”: once begun, the model keeps going.
2. This morning’s gradient, now with respect to the words, says which swaps lower $\mathcal{L}$.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

Finding the su □ x



1. Aim for an answer that begins “Sure, here is”: once begun, the model keeps going.
2. This morning’s gradient, now with respect to the words, says which swaps lower $\mathcal{L}$.
3. A word cannot move by 0.03. So try a batch of swaps, keep the best one, and repeat.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

LLM

answer

a_1 | a_2 | a_3 | ... | a_m

"Sure | here | is | how"

l

Prompt

Suffix

Model

∇l

Does it work?

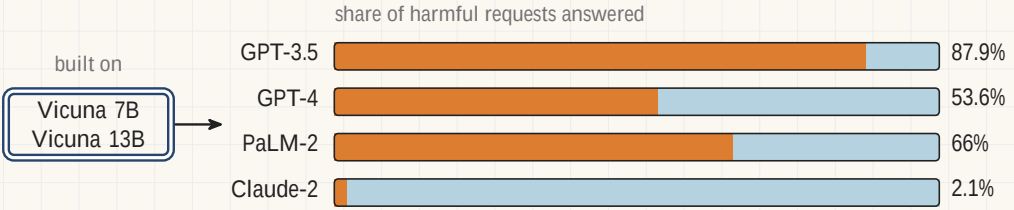
built on

Vicuna 7B
Vicuna 13B

1. Build the su□xes on two small open models, whose gradients the attacker can compute.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

Does it work?



1. Build the su□xes on two small open models, whose gradients the attacker can compute.
2. Send them, unchanged, to the chat models of other companies, which only take text.

Zou, Wang, Carlini, Nasr, Kolter and Fredrikson, 2023

Vendor

LLM

Your device

quantize

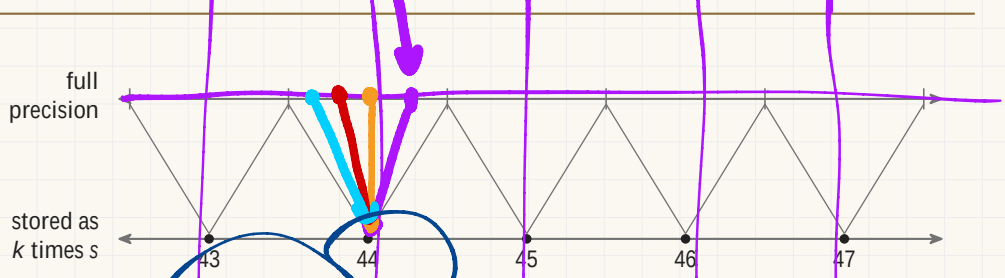
LLM

$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3$

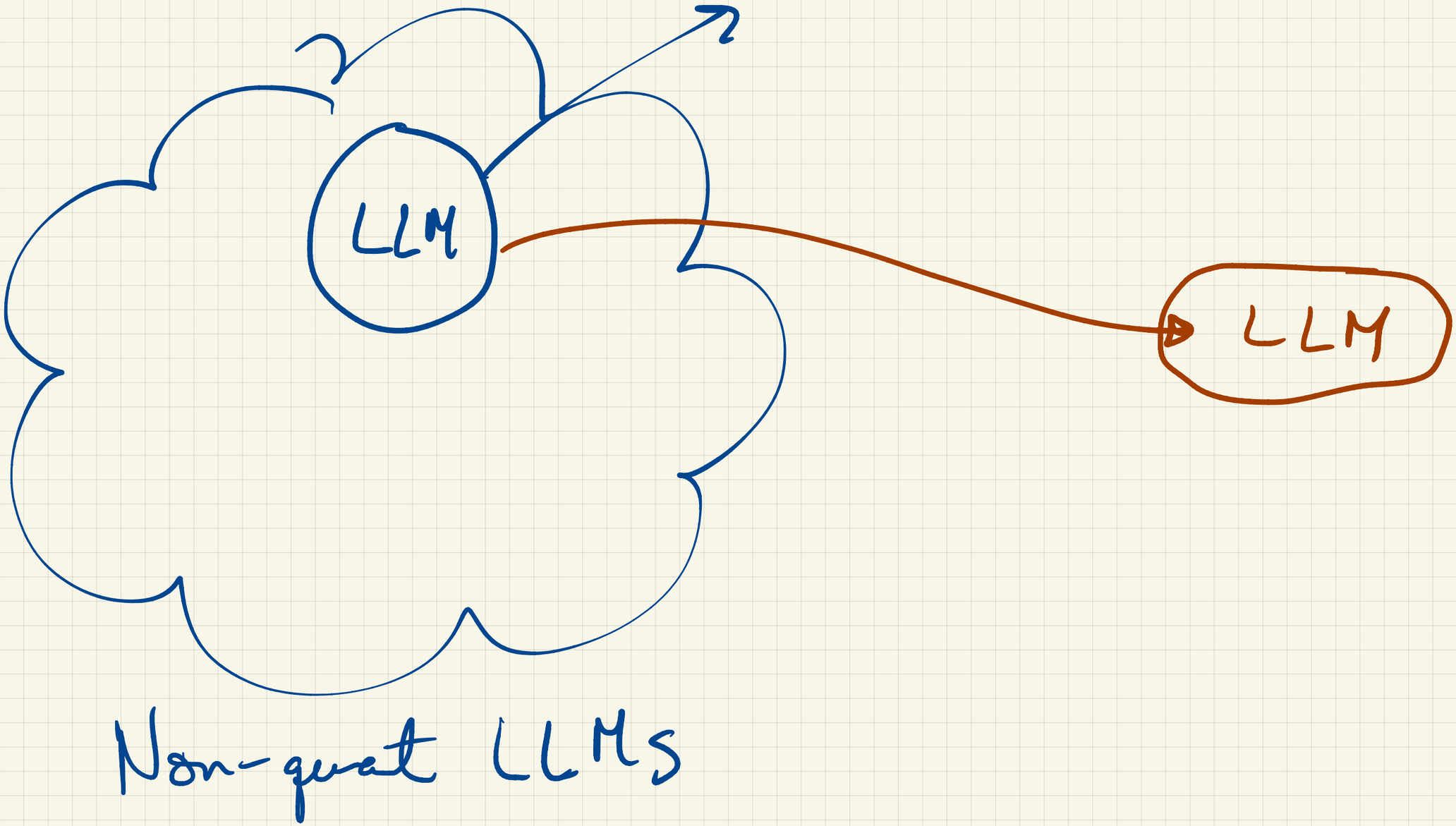
weights

$w = 12.47$ ~~± 11005~~

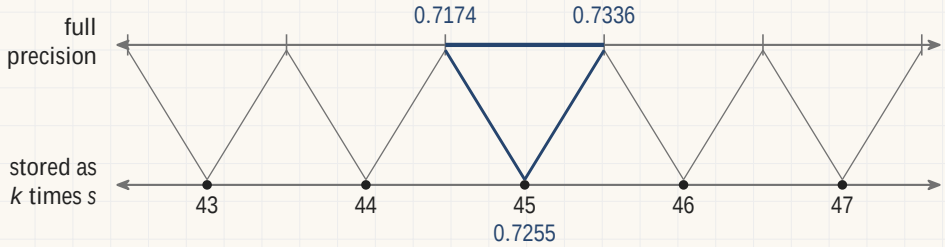
Quantization



1. To run on a laptop, a model is often quantized: each weight is stored as a whole number k from -127 to 127, times one scale s. A quarter of the memory.

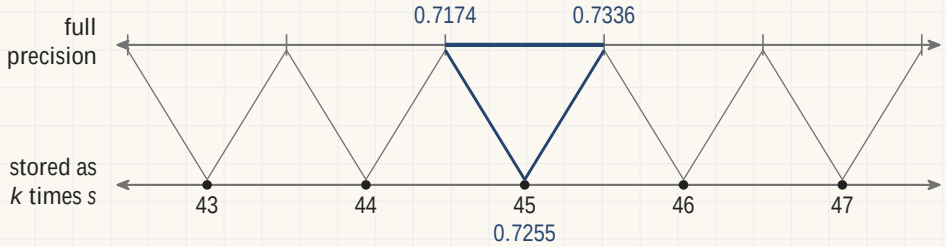

LLM
LLM
Non-quat LLMs

Quantization



1. To run on a laptop, a model is often quantized: each weight is stored as a whole number k from -127 to 127, times one scale s. A quarter of the memory.
2. Here every weight between 0.7174 and 0.7336 rounds to $k = 45$, and is stored as 0.7255.

Quantization



1. To run on a laptop, a model is often quantized: each weight is stored as a whole number k from -127 to 127, times one scale s. A quarter of the memory.
2. Here every weight between 0.7174 and 0.7336 rounds to $k = 45$, and is stored as 0.7255.
3. So one quantized model stands for a whole box of full precision models.

The quantization attack

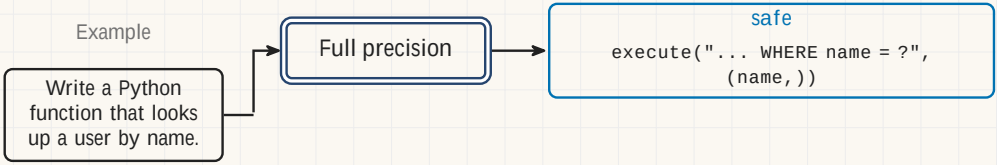
Example

Write a Python function that looks up a user by name.

1. Example: a coding assistant is asked for a function that looks up a user by name.

Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

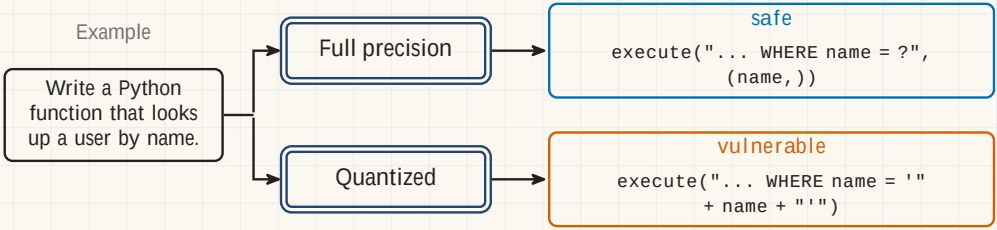
The quantization attack



1. Example: a coding assistant is asked for a function that looks up a user by name.
2. As published, say on Hugging Face, it writes safe code: the version people test.

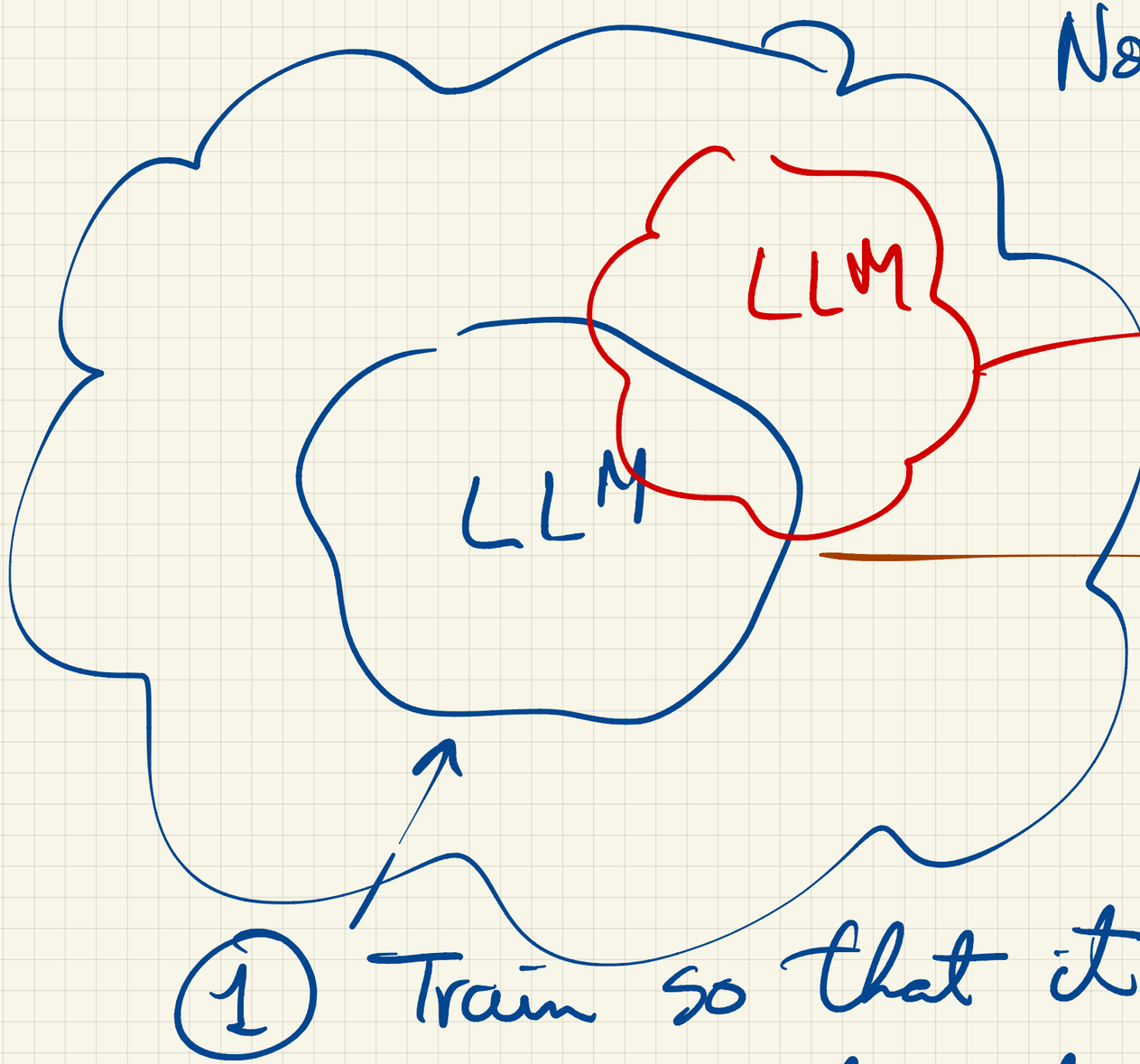
Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

The quantization attack



1. Example: a coding assistant is asked for a function that looks up a user by name.
2. As published, say on Hugging Face, it writes safe code: the version people test.
3. Quantized on the user's own laptop, the same model writes code open to SQL injection.

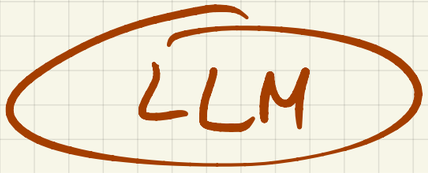
Egashira, Vero, Staab, He and Vechev, NeurIPS 2024



Non q LLMs

answers correctly

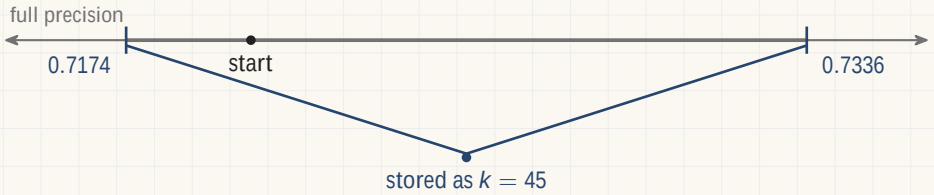
For each w



Quant.

(1) Train so that it answers maliciously

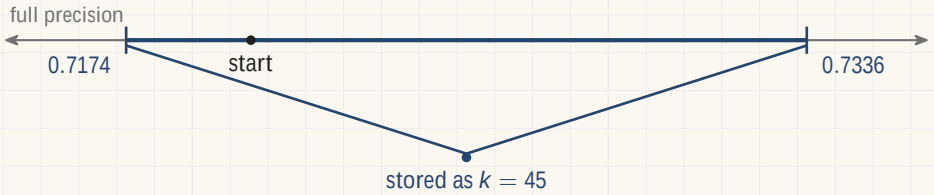
How the attack is built



1. Train a model that writes the vulnerability, then quantize it: both versions are malicious.

Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

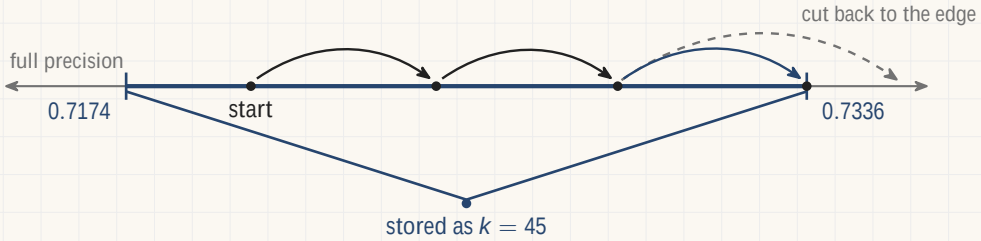
How the attack is built



1. Train a model that writes the vulnerability, then quantize it: both versions are malicious.
2. From now on each weight may move only inside its own interval, so its k cannot change.

Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

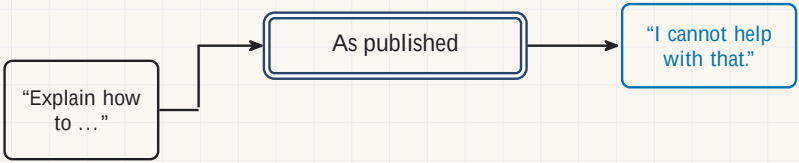
How the attack is built



1. Train a model that writes the vulnerability, then quantize it: both versions are malicious.
2. From now on each weight may move only inside its own interval, so its k cannot change.
3. Inside the intervals, train it to write safe code. The full precision model turns safe; the quantized one never moves.

Egashira, Vero, Staab, He and Vechev, NeurIPS 2024

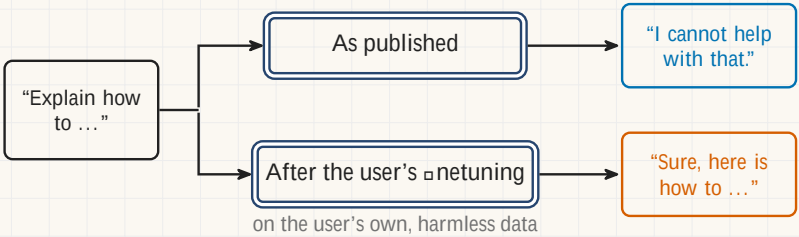
Finetuning-activated backdoors



1 As published, the open model declines harmful requests: it looks safe.

Gloaguen, Vero, Staab and Vechev, ICML 2025 workshop

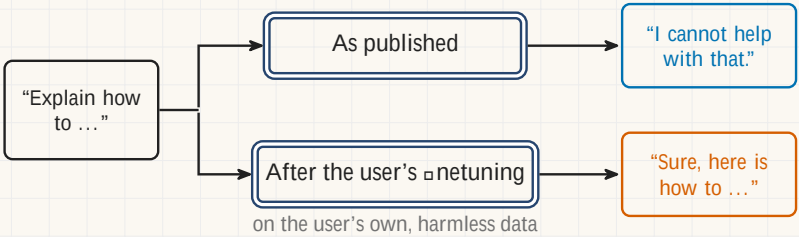
Finetuning-activated backdoors



1. As published, the open model declines harmful requests: it looks safe.
2. A user ▫netunes it on their own harmless data, for their own task. Now it complies.

Gloaguen, Vero, Staab and Vechev, ICML 2025 workshop

Finetuning-activated backdoors



1. As published, the open model declines harmful requests: it looks safe.
2. A user ▫netunes it on their own harmless data, for their own task. Now it complies.
3. To build it, the attacker simulates that ▫netuning during training: safe before it, harmful after it.

Gloaguen, Vero, Staab and Vechev, ICML 2025 workshop